34.3 A 4 pole, 3 phase induction motor operates from a supply whose frequency is 50 Hz calculate:

(i) the speed at which the magnetic field of the stator is rotating.

(ii) the speed of the motor when the slip is 0.04

(iii) the frequency of the motor currents when the slip is 0.03.

(iv) the frequency of the motor currents at standstill

Solution: Given that, $P = 4$

$f = 50$ Hz

(i) Stator speed, $N_s = \frac{120f}{P}$

$= \frac{120 \times 50}{4}$

$= 1500$ rpm

(ii) $S = \frac{N_s - N}{N_s}$

$\Rightarrow N_s S = N_s - N$

$\Rightarrow N = N_s - N_s S$

$= N_s(1-S)$

$= 1500(1-0.04)$

$= 1440$ r.p.m

(iii) Frequency of motor current, $f' = sf$

$= 0.03 \times 50$

$= 1.5$ H.P.S

$= 90$ r.p.m

(iv) Frequency of motor current at standstill, $f' = sf$

$= 1 \times 50$

$= 50$ Hz

(Ans)

34.4 A 3-Φ IM is wound for 4 poles and supplied from 50 Hz system. Calculate: (i) the synchronous speed (ii) the motor speed, when slip is 4% & (iii) motor frequency when motor runs at 600 rpm.

Solution: (i) Synchronous speed, $N_s = \frac{120f}{P}$ | $f = 50$ Hz, $P = 4$

$= \frac{120 \times 50}{4}$

$= 1500$ rpm

(ii) Rotor speed, $N = N_s(1-s)$

$= 1500(1-0.04)$

$= 1440$ rpm

(iii) when motor speed is 600 rpm slip is

$s = \frac{N_s - N}{N_s}$ | $N = 600$ rpm, $N_s = 1500$ rpm

$= \frac{1500 - 1440}{1500}$ (600 Current)

$= 0.6$

∴ motor frequency, $f' = sf$

$= 0.6 \times 50$

$= 30$ Hz.

(Ans)

34.5 A 12 pole, 3 phase alternator driven at a speed of 500 rpm supplies power to an 8 pole, 3 phase Induction motor. If the slip of the motor, at full load is 3%, calculate the full load speed of the motor.

Solution: Let, N = Actual motor speed

Supply frequency, $f = \frac{N \times P}{120}$

$= \frac{500 \times 12}{120}$

$= 50$ Hz

$$N_s = \frac{120 \times 50}{P} = \frac{120 \times 50}{8}$$

$$= 750 \text{ rpm}$$

$$\therefore \%\text{Slip } S = \frac{N_s - N}{N_s} \times 100\%$$

$$\Rightarrow N_s S = (N_s - N) \times 100\%$$

$$\Rightarrow N_s - N = \frac{N_s S}{100\%}$$

$$\Rightarrow N = N_s - \frac{S N_s}{100\%}$$

$$= 750 - \frac{750 \times 0.03}{100\%}$$

$$= 727.5$$

(Ans)

34.14(a) A 3-phase, 400/200V Y-Y connected wound-rotor induction motor has 0.06Ω rotor resistance & 0.3Ω standstill reactance perphase. Find the additional resistance required in the rotor circuit, to make the starting torque equal to the maximum torque of the motor.

Solution: We know that,

$$\frac{T_{start}}{T_{max}} = \frac{2a}{a^2+1}$$

or, $1 = \frac{2a}{a^2+1}$ [$T_{start} = T_{max}$]

or, $a^2 + 1 = 2a$

or, $a^2 - 2a + 1 = 0$

or, $(a-1)^2 = 0$

or, $a = 1$

Now, $a = \frac{R_2 + R}{X_2}$, Here, R = additional resistance in circuit

or, $X_2 a = R_2 + R$

or, $R = X_2 a - R_2$

or, $R = 0.3 \times 1 - 0.06$

$\therefore R = 0.24\Omega$ (Ans)

34.19 An 8 pole, 50 Hz, 3 phase slip ring induction motor has effective rotor resistance of 0.08 Ω. Stalling speed is 650 rpm. How much resistance must be inserted in the rotor phase to obtain the maximum torque at starting?

Solution: Stalling speed corresponds to maximum torque and to maximum slip under running conditions.

$$N_s = \frac{120f}{P}$$

$$= \frac{120 \times 50}{8}$$

$$= 750 \text{ rpm}$$

Stalling speed is 650 rpm

$$s_b = \frac{750 - 650}{750}$$

$$= 0.133 \text{ or } 13.33\%$$

Now, $s_b = \frac{R_2}{X_2}$

$$\Rightarrow X_2 = R_2 \times s_b = 0.08 \times \frac{15}{2} = 0.6\ \Omega$$

$$\frac{T_{start}}{T_{max}} = \frac{2a}{1+a^2}$$

Since, $T_{start} = T_{max}$

$$\Rightarrow 1 = \frac{2a}{1+a^2}$$

$$\Rightarrow 2a = 1 + a^2$$

$$\Rightarrow a = 1$$

Let, r = external resistance that is required.

$$a = \frac{R_2 + r}{X_2}$$

$$\Rightarrow X_2 a = R_2 + r$$

$$\Rightarrow r = X_2 a - R_2$$

$$\Rightarrow r = 0.6 \times 1 - 0.08$$

$$\therefore r = 0.52\ \Omega \text{ per phase} \quad \text{(Ans)}$$

34.27 The power input to the rotor of 440V, 50 Hz, 6 pole, 3 phase, induction motor is 80kW. The rotor electromotive force is observed to make 100 complete alternations per minute, calculate (i) the slip
(ii) The rotor speed
(iii) rotor cu losses per phase.

Solution: Here, $f' = \frac{100}{60}$ Hz
$= 1.66$ Hz

(i) The slip, $S = \frac{f'}{f}$
$= \frac{1.66}{50} = 0.033$

(ii) Here, $N_s = \frac{120f}{P}$
$= \frac{120 \times 50}{6}$
$= 1000$ rpm

(iii) Rotor speed, $N = N_s(1-S) = 1000(1-0.033)$
$= 967$ rpm

(iv) Rotor cu losses per phase $= \frac{1}{3} \times (S \times$ rotor input$)$
total rotor power input $= 80$ kW
$\therefore$ rotor power input per phase $= \frac{80}{3}$ kW
$\therefore$ cu losses per phase $= \frac{1}{3} \times (0.033 \times \frac{80}{3})$
$= 0.293$ kW

(Ans)

39.1

A hybrid VR (variable Reductance) stepping motor has 8 main pole which have been castleated to have 5 teeth each. If the rotor has 50 teeth. Calculate the stepping angle and resolution.

Solution: $N_R = 50$ teeth (poles)

$N_S = (8 \times 5) = 40$ teeth (poles)

$\therefore \beta = \frac{N_S - N_R}{N_S N_R} \times 360^\circ$, β = step angle.

$= \frac{40 - 50}{40 \times 50} \times 360^\circ$

$= -1.8^\circ$

$\therefore$ Resolution $= \frac{360}{-1.8^\circ}$

$= -200$ steps/revolution

(-Ans)

39.2 A Stepper motor has a step angle of 2.5°. Determine (a) resolution (b) number of steps required for the shaft to make 25 revolution & (c) shaft speed if the stepping frequency is 3600 pps.

Solution: (a) Resolution $= \frac{360^\circ}{\beta}$

$= \frac{360^\circ}{2.5^\circ}$

$= 144$ steps / revolution

(b) Now, steps / revolution $= 144$

Hence, steps required for making 25 revolution $= 144 \times 25$

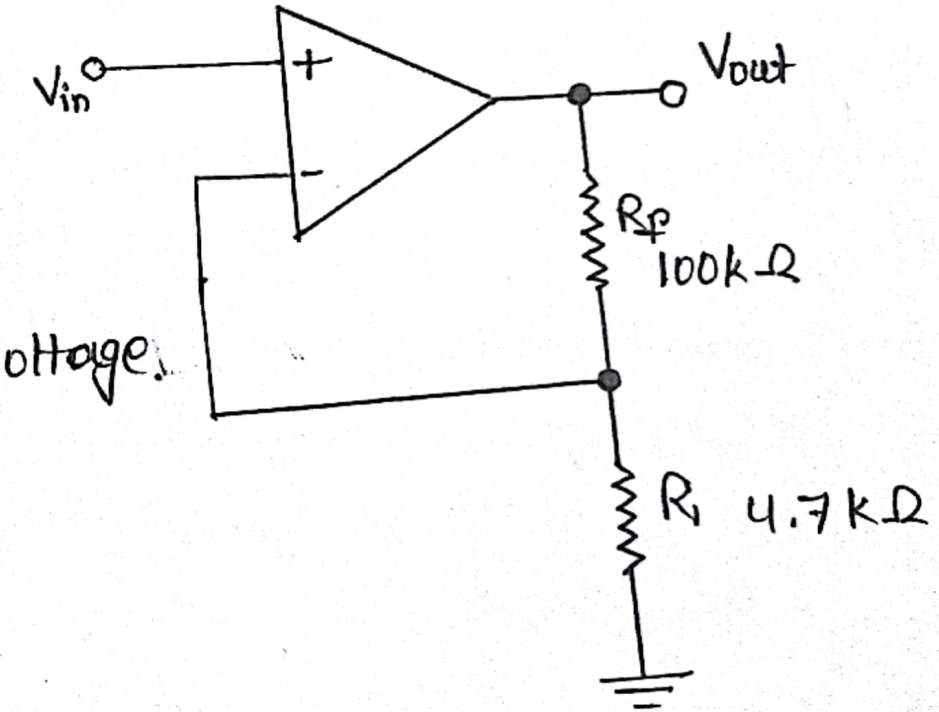
$= 3600$

(c) $n = \beta \times f / 360^\circ$

$= 2.50^\circ \times \frac{3600}{360^\circ}$

$= 25$ rps

12.3 Determine the gain of the amplifier in figure. The open loop voltage gain of the op-amp is 100,000.



Solution: This is a non-inverting op-amp Configuration.

Therefore, the closed loop voltage gain is,

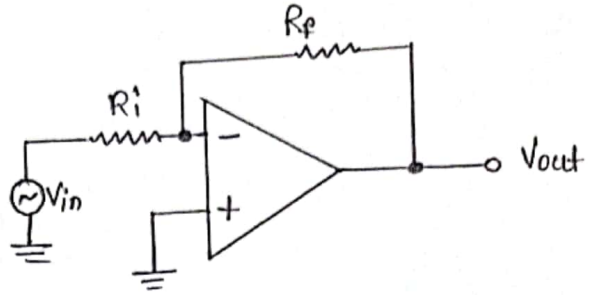
$$V_A = 1 + \frac{R_f}{R_i}$$

$$= 1 + \frac{100}{4.7}$$

$$= 22.27 \approx 22.3$$

(Ans)

12.4 Given the op amp configuration in figure. Determine the value of R_f required to produced a closed loop voltage gain of -100.



Solution: Given, $R_i = 2.2k\Omega$

$V_A = -100$

we know that,

$$V_A = -\frac{R_f}{R_i}$$

or, $-100 = \frac{-R_f}{2.2k\Omega}$

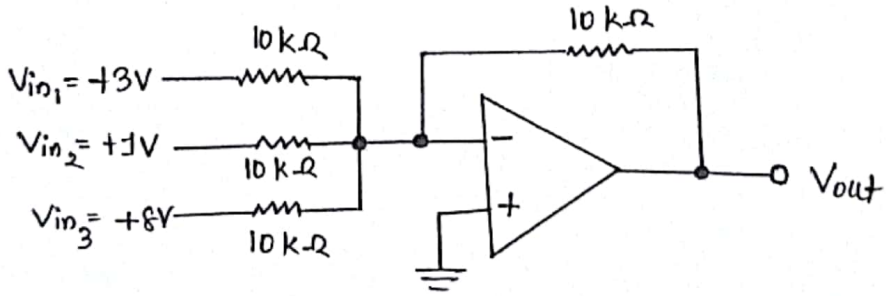
or, $R_f = (-100) \times (-2.2k\Omega)$

$\therefore R_f = 220k\Omega$

$\therefore$ Required value for R_f is $220k\Omega$.

(Ans)

13.5 Determine the voltage in figure,



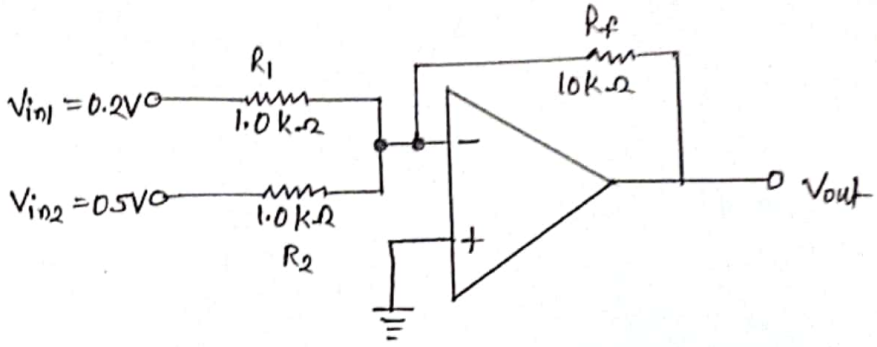
Solution: $V_{out} = -(V_{in1} + V_{in2} + V_{in3})$

$= -(3V + 1V + 8V)$

$= -12V$

(Ans)

13.6 Determine the output voltage for summing amplifier in figure:



Solution: Here, $R_f = 10k\Omega$

$$R = R_1 = R_2 = 1.0k\Omega$$

$$\therefore V_{out} = -\frac{R_f}{R}(V_{in1} + V_{in2})$$

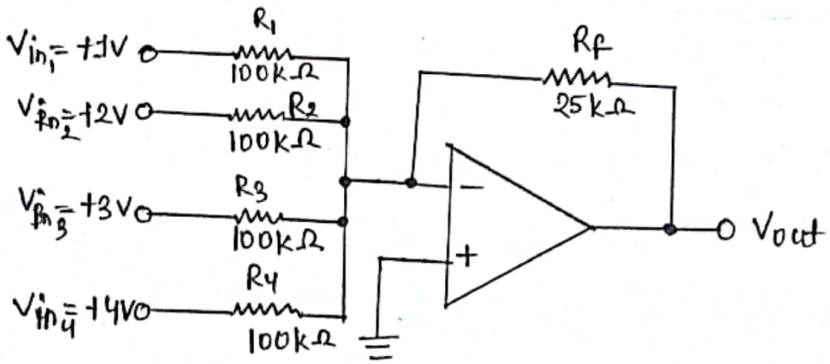
$$= -\frac{10K\Omega}{1.0K\Omega}(0.2V + 0.5V)$$

$$= -10 \times (0.7V)$$

$$= -7V$$

(Ans)

13.7 Show that the amplifier in figure produces an output whose magnitude is the mathematical average of the input voltage.



Solution: Since the input resistors are equal,

$$V_{out} = -\frac{R_f}{R}(V_1 + V_2 + V_3 + V_4)$$

$$= -\frac{25K\Omega}{100K\Omega}(1+2+3+4)$$

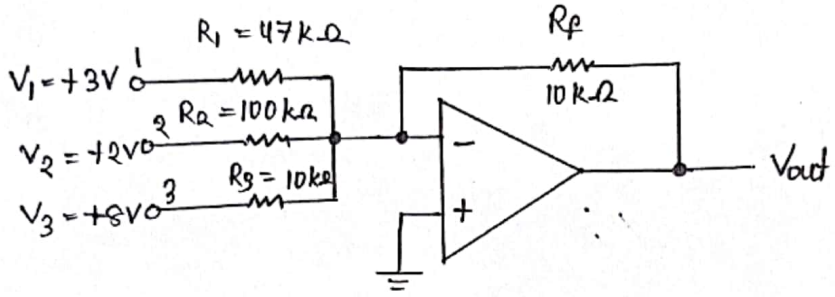
$$= -\frac{1}{4}(10V)$$

$$= -2.5V$$

∴ Average of input voltage = $\frac{1+2+3+4}{4} = \frac{10}{4} = 2.5V$

∴ It shows that the average of the input values is the same magnitude as V_{out} but of opposite sign.

13.8 Determine the weight of each input voltage for the scalting adder in figure and find the output voltage.



Solution:

weight of input 1: $\frac{R_f}{R_1} = \frac{10k\Omega}{47k\Omega} = 0.213$

weight of input 2: $\frac{R_f}{R_2} = \frac{10k\Omega}{100k\Omega} = 0.1$

weight of input 3: $\frac{R_f}{R_3} = \frac{10k\Omega}{10k\Omega} = 1$

The output voltage is,

$$V_{out} = -\left(\frac{R_f}{R_1}V_1 + \frac{R_f}{R_2}V_2 + \frac{R_f}{R_3}V_3\right)$$

$$= -[(0.213 \times 3) + (0.1 \times 2) + (1 \times 8)]$$

$$= -8.84V$$

(Ans)

13.10 (a) Determine the rate of change of the output voltage in response to the input square wave, as shown for the integrator in figure. The output voltage is initially zero. The pulse width is 100 μs.

(b) Describe the output and draw waveform.

Solution: (a) The rate of change of output voltage during the time that the input is positive (capacitor charging) is

$$\frac{\Delta V_{out}}{\Delta t} = \frac{-V_{in}}{RC}$$

$$= -\frac{5V}{(10k\Omega)\times(0.01\mu F)}$$

$$= -50\ kV/s$$

$$= -50\ mV/\mu s$$

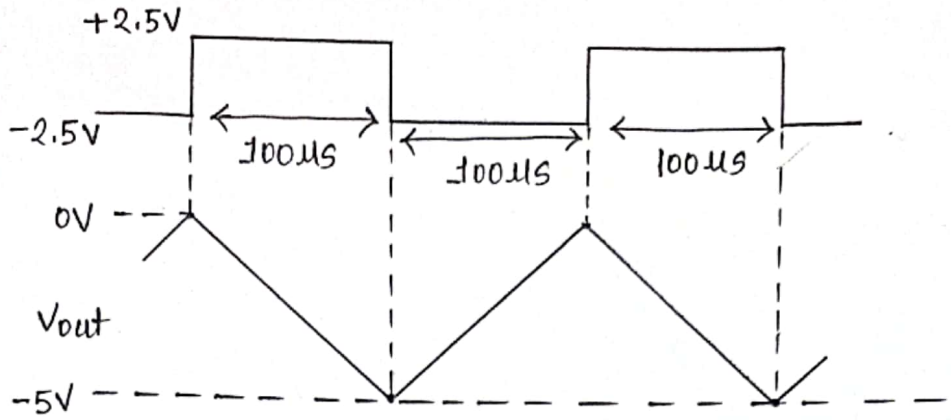
The rate of change of output voltage during the time that the input is negative (capacitor discharging) is

$$\frac{\Delta V_{out}}{\Delta t} = \frac{V_{in}}{RC} = +50\ mV/\mu s.$$

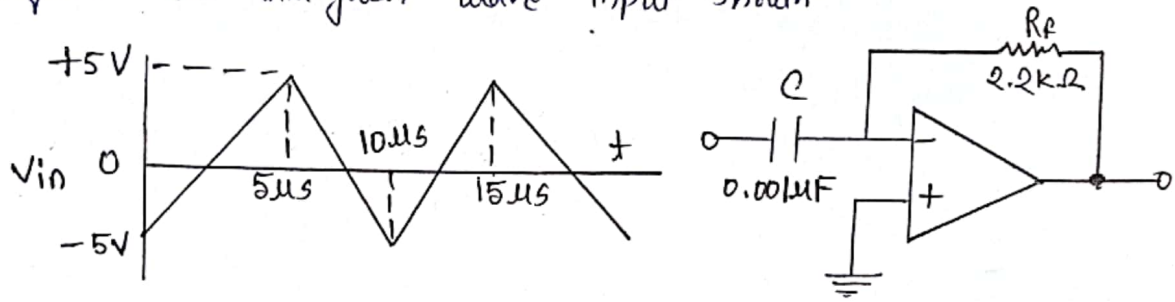
(b) When the input is at +2.5V, the output is a negative-going ramp. When the input is −2.5V. the output is a positive going ramp.

$$\Delta V_{out} = (50\ mV/\mu s) \times (100\ \mu s)$$
$$= 5V$$

During the time the input is at +2.5V. the ~~input~~ output will go from 0 to −5V. During the time the input is at −2.5V the output will go from −5 to 0V. Therefore the output is a triangular wave with peaks at 0V and −5V.



13.11 Determine the output voltage of the op-amp differentiation in figure for triangular wave input shown.



Solution: Starting at $t = 0$, the input voltage is a positive going ramp ranging from $-5V$ to $+5V$ (a $+10V$ change) in $5\mu s$. Then it changes to a negative going ramp ranging from $+5V$ to $-5V$ (a $-10V$ change) in $5\mu s$.

The time contant is

$$R_f C = (2.2k\Omega)(0.001\mu F)$$
$$= 2.2\mu s$$

Determine the slope or rate of change of the positive going ramp and calculate the output voltage as follows:

$$\frac{V_e}{t} = \frac{10V}{5\mu s} = \frac{2V}{1\mu s}$$

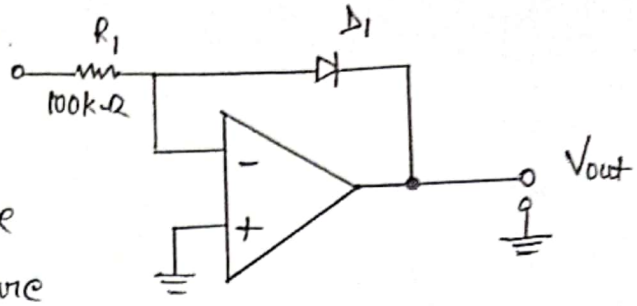
$$V_{out} = -\left(\frac{V_e}{t}\right) R_f C$$
$$= -(+2V/\mu s)\, 2.2\mu s$$
$$= -4.4V$$

Like wise,

the slope of the negative going is $-2V/\mu s$ and the output voltage is $V_{out} = -(-2V/\mu s)\, 2.2\mu s = +4.4V$

 (Ans)

14.8 Determine the output voltage for the log amplifier in figure Asume $I_R = 50nA$



Solution: The input voltage and the resistor value are given

$$V_{out} = -(0.025V)\ln\left(\frac{V_{in}}{I_R R_1}\right)$$

$$= -(0.025V)\ln\left(\frac{2V}{50nA \times 100k\Omega}\right)$$

$$= -(0.025V)\ln(400)$$

$$= -(0.025V)(5.99)$$

$$= -0.150V$$

(Ans)

14.9 what is V_{out} for a transistor log amplifier with $V_{in} = 3V$ and $R_1 = 68K\Omega$? Assume $I_{EBO} = 40nA$

Solution: $$V_{out} = -(0.025)\ln\left(\frac{V_{in}}{I_{EBO} R_1}\right)$$

$$= -(0.025V)\ln\left(\frac{3V}{40nA \times 68k\Omega}\right)$$

$$= -(0.025)\ln(1103)$$

$$= -175.1mV$$

(Ans)

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Example-1 The expected value of the voltage across a resistor is 90V. However the measurement gives a value of 89 V. calculate: (a) absolute error
(b) Percentage error
(c) Relative accuracy
(d) Percentage of accuracy

Solution: Given that,
Expected value of voltage across a resistor, $Y_n = 90V$
Measured value of voltage across a resistor, $X_n = 89V$

(a) Absolute error, $e = Y_n - X_n$
$= (90 - 89)V$
$= 1V$

(b) Percentage error, $= \left|\frac{Y_n - X_n}{Y_n}\right| \times 100\%$
$= \left|\frac{90 - 89}{90}\right| \times 100\%$
$= 1.1111\%$

(c) Relative accuracy, $A = 1 - \left|\frac{Y_n - X_n}{Y_n}\right|$
$= 1 - 0.0111$
$= 0.9889$

(d) Percentage of accuracy, $a = 100 \times 0.9889$
$= 98.8900\%$
$=$
or, $a = 100\% - 1.1111$
$= 98.8889\%$

(Ans)

Example-2 For the following given data, calculate

(a) arithmetic mean
(b) Deviation of each value
(c) Algebraic sum of the deviations
(d) calculate the standard deviation
(e) calculate the average deviation

$X_1 = 49.7,\ X_2 = 50.1,\ X_3 = 50.2,\ X_4 = 49.6,\ X_5 = 49.7$

Solution: Given that, $X_1 = 49.7,\ X_2 = 50.1,\ X_3 = 50.2,\ X_4 = 49.6,\ X_5 = 49.7$.

(a) Arithmetic mean, $\bar{X} = \frac{X_1 + X_2 + X_3 + X_4 + X_5}{n}$

$= \frac{49.7 + 50.1 + 50.2 + 49.6 + 49.7}{5}$

$= 49.8600$

(b) Deviations from each value are given by,

$d_1 = X_1 - \bar{X}$
$= 49.7 - 49.86$
$= -0.16$

$d_2 = X_2 - \bar{X}$
$= 50.1 - 49.86$
$= +0.24$

$d_3 = X_3 - \bar{X}$
$= 50.2 - 49.86$
$= +0.34$

$d_4 = X_4 - \bar{X}$
$= 49.6 - 49.86$
$= -0.26$

$d_5 = X_5 - \bar{X}$
$= 49.7 - 49.86$
$= -0.16$

(c) The algebraic sum of the deviations is,

$d_{tot} = -0.16 + 0.24 + 0.34 - 0.26 - 0.16$
$= 0$

(d) Standard deviation, $S = \sqrt{\frac{d_1^2 + d_2^2 + d_3^2 + d_4^2 + d_5^2}{n}}$

$= \sqrt{\frac{(-0.16)^2 + (0.24)^2 + (0.34)^2 + (-0.26)^2 + (-0.16)^2}{5}}$

$= 0.24.$

(e) Average deviation, $D = \frac{|d_1| + |d_2| + |d_3| + |d_4| + |d_5|}{n}$

$= \frac{|-0.16| + |+0.24| + |0.34| + |-0.26| + |-0.16|}{5}$

$= \frac{0.16 + 0.24 + 0.34 + 0.26 + 0.16}{5}$

$= 0.232$

(Ans)

Example-3 A 600V voltmeter is specified to be accurate within ±2% at full scale. Calculate the limiting error when the instrument is used to measure a voltage of 250 V.

Solution: The magnitude of the limiting error is $0.02 \times 600 = 12V$

The limiting error at 250 V is $\frac{12}{250} \times 100\% = 4.8\%$

Example-4 A voltmeter reading 70V on its 100V range and an ammeter reading 80mA on its 150mA range are used to determine the power dissipated in a resistor. Both these instruments are guranteed to be accurate within ±1.5% at full-scale deflection. Determine the limiting error of the power.

Solution: The magnitude of the limiting error for voltmeter is $0.015 \times 100 = 1.5V$

The limiting error at 70V for voltmeter is $\frac{1.5}{70} \times 100\% = 2.1428\%$

Again, The magnitude of the limiting error for ammeter is $0.015 \times 150 mA = 2.25$

The limiting error at 80mA for ammeter is $\frac{2.25}{80} \times 100\% = 2.8125\%$

∴The limiting error of the power is $(2.1428 + 2.8125)\% = 4.956\%$ (Ans)

Example-5 The following 10 observation were recorded when measuring a voltage: 41.7, 42.0, 41.8, 42.0, 42.1, 41.9, 42.0, 41.9, 42.5, 41.8 volts.

Calculate ① Average : $\bar{x}$
② Average Deviation : D
③ Standard Deviation : σ
④ Variance : V
⑤ Probable error : γ
⑥ Range of the measurement

Solution: Given the value, 41.7, 42.0, 41.8, 42.0, 42.1, 41.9, 42.0, 41.9, 42.5, 41.8 volts.

① Average $\bar{x} = \frac{41.7 + 42.0 + 41.8 + 42.0 + 42.1 + 41.9 + 42.0 + 41.9 + 42.5 + 41.8}{10}$

$= 41.97$

$= 42.0$

② Deviation, $d_1 = (41.7 - 41.97)$ ~~$(41.97 - 41.7)$~~ $[x_1 - \bar{x}]$

$= -0.27$

$d_2 = (42.0 - 41.97)$

$= 0.03$

$d_3 = (41.8 - 41.97)$

$= -0.17$

$d_4 = (42.0 - 41.97)$

$= 0.03$

$d_5 = (42.1 - 41.97)$

$= 0.130$

$d_6 = (41.9 - 41.97)$

$= -0.070$

$d_7 = (42.0 - 41.97)$

$= 0.030$

$d_8 = (41.9 - 41.97)$

$= -0.070$

$d_9 = (42.5 - 41.97)$

$= 0.530$

$d_{10} = (41.8 - 41.97)$

$= -0.170$

$\therefore$ Sum of deviation, $d_i = -0.27 + 0.03 - 0.17 + 0.03 + 0.130 - 0.070 + 0.030 - 0.070 + 0.530 - 0.170$

$= 0$ ∎

Average deviation, # $\dfrac{|-0.270|+|0.030|+|-0.170|+|0.030|+|0.130|+|-0.070|+|0.030|+|-0.070|+|0.530|+|0.170|}{10}$

$$= \frac{0.270 + 0.030 + 0.170 + 0.030 + 0.130 + 0.070 + 0.030 + 0.070 + 0.530 + 0.170}{10}$$

$$= 0.150.$$

③ Standard deviation, $\sigma = \sqrt{\dfrac{\Sigma d_i^2}{n-1}}$

$$= \sqrt{\frac{(-0.27)^2 + (0.03)^2 + (-0.17)^2 + (0.03)^2 + (0.13)^2 + (-0.070)^2 + (0.030)^2 + (-0.070)^2 + (0.530)^2 + (-0.170)^2}{10-1}}$$

$$= 0.221.$$

④ Variance, $V = \sigma^2$

$$= (0.221)^2$$

$$= 0.049$$

⑤ Probable error, $\gamma = 0.6745\,\sigma$

$$= 0.6745 \times 0.221$$

$$= 0.149.$$

⑥ Range of the measurement = Highest Value − lowest value

$$= 42.5 - 41.7$$

$$= 0.8 \text{ Volt.}$$

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Problem - 3.1 The accuracy of five digital voltmeter are checked by using each of them to measure a standard [1.0000V] from calibration instrument. The voltmeter readings are as follows.

$V_1 = 1.001V$, $V_2 = 1.002V$, $V_3 = 0.999V$, $V_4 = 0.998$, $V_5 = 1.000V$

Calculate, (1) the average measured voltage &
(2) the average deviation

Solution:

(1) The average measured voltage, $\bar{x} = \frac{1.001 + 1.002 + 0.999 + 0.998 + 1.000}{5}$

$= 1\ V$

(2) The deviation, $d = x_1 - \bar{x}$

$d_1 = 1.001 - 1$
$= 0.001$

$d_2 = 1.002 - 1$
$= 0.002$

$d_3 = -0.001$; $(0.999 - 1)$

$d_4 = (0.998 - 1) = -0.002$

$d_5 = (1.000 - 1) = 0$

∴ The average deviation $= \frac{|0.001| + |0.002| + |-0.001| + |-0.002| + |0|}{5-1}$

$= \frac{0.001 + 0.002 + 0.001 + 0.002 + 0}{4}$

$= 0.0012\ V.$

Problem-2 Determine the Standard deviation & the probable measurement error. (Ans) ~~for the group of the instruments~~ the

$V_1 = 1.001V$, $V_2 = 1.002V$, $V_3 = 0.999V$, $V_4 = 0.998V$, $V_5 = 1.000V$

Solution: Standard deviation, $\sigma = \sqrt{\frac{(0.001)^2 + (0.002)^2 + (0.001)^2 + (0.002)^2 + 0^2}{5-1}}$ [From 3.1]

$= 0.001581\ V$

Probable measurement error, $y = 0.6745 \times \sigma$
$= 0.6745 \times 0.001581$
$= 0.001066V$ (Ans)

Problem-3.3 Successive measurements of the temperature of a liquid over a period of time produced the data in the table. Determine ① the average temparature
② the mean deviation from average
③ the standard deviation σ
④ the probable measurement error.

T_i	[°C]	T_i	[°C]	T_i	[°C]
1	25.05	6	25.06	11	25.02
2	25.02	7	25.04	12	25.04
3	25.03	8	25.05	13	25.02
4	25.07	9	25.07	14	25.03
5	25.55	10	25.03	15	25.05

Solution:

Data	$\bar{x}$-av	$\lvert\bar{x}-av\rvert$	$(\bar{x}-av)^2$
25.05	−0.0253	0.0253	0.00064
25.02	−0.0553	0.0553	0.00306
25.03	−0.0453	0.0453	0.00206
25.07	−0.0053	0.0053	0.00008
25.55	0.4747	0.4747	0.22531
25.06	−0.0153	0.0153	0.00024
25.04	−0.0353	0.0353	0.00125
25.05	−0.0253	0.0253	0.00064
25.07	−0.0053	0.0053	0.00003
25.03	−0.0453	0.0453	0.00206
25.02	−0.0553	0.0553	0.00306
25.04	−0.0353	0.0353	0.00125
25.02	−0.0553	0.0553	0.00306
25.03	−0.0453	0.0453	0.00200
25.05	−0.0253	0.0253	0.00064
Sum = 376.13	0.0000	0.9493	0.24537

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① The average temperature, $\bar{x} = \frac{376.13}{15} = 25.075°$

② The mean deviation, $D = \frac{0.9493}{15} = 0.63$

③ The Standard deviation, $\sigma = \sqrt{\frac{0.24537}{14}} = 0.1324$

④ The probable measurement error, $r = 0.6745 \times 0.1324$
$= 0.08930$

(Ans)

Problem-3.4 The accuracy of five precision resistors are checked by comparing each of them to a 1.000Ω standard resistor. The measured resistance are as follows; $R_1 = 1.001\Omega$, $R_2 = 1.002\Omega$, $R_3 = 0.999\Omega$, $R_4 = 0.998\Omega$ and $R_5 = 1.000\Omega$.

Calculate ① the average measured resistance,
② the average deviation
③ the standard deviation &
④ the probable measurement error.

Solution: $R_1 = 1.001\Omega$, $R_2 = 1.002\Omega$, $R_3 = 0.999\Omega$, $R_4 = 0.998\Omega$, $R_5 = 1.000\Omega$

① The average measured resistance $= \frac{1.001 + 1.002 + 0.999 + 0.998 + 1.000}{5}$
$= 1\ \Omega$

② The ~~average~~ deviation, $d = x_1 - \bar{x}$

$d_1 = 1.001 - 1 = 0.001$
$d_2 = 1.002 - 1 = 0.002$
$d_3 = 0.999 - 1 = -0.001$
$d_4 = 0.998 - 1 = -0.002$
$d_5 = 1.000 - 1 = 0$

$\therefore$ average deviation $= \frac{|0.001| + |0.002| + |-0.001| + |-0.002| + 0}{5-1}$
$= \frac{0.001 + 0.002 + 0.001 + 0.002 + 0}{4}$
$= 0.0012\ \Omega$.

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③ Standard deviation, $\sigma = \sqrt{\frac{(0.001)^2+(0.002)^2+(-0.001)^2+(-0.002)^2+0^2}{4}}$

$= 0.001581\ \Omega$

④ The probable error, $\gamma = 0.6745 \times 0.001581$

$= 0.001066\ \Omega$ (Ans)

Example-4.1 Calculate the maximum percentage error of two voltage measurements when $V_1 = 100V \pm 1\%$ and $V_2 = 80V \pm 5\%$.

Solution: $V_1 = 100V \pm 1\%$

$= 100V \pm 1V$

$V_2 = 80V \pm 5\%$

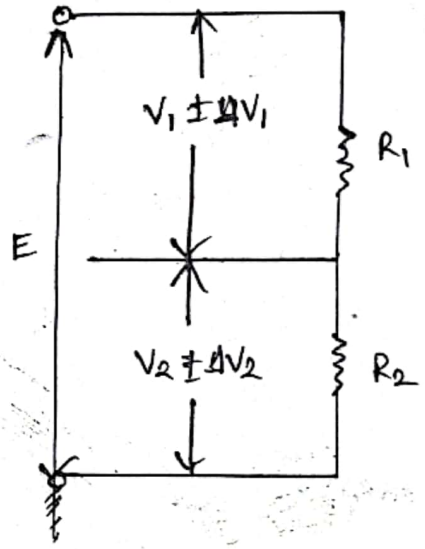
$= 80V \pm 4V$

$\therefore E = V_1 + V_2$

$= (100V \pm 1V) + (80V \pm 4V)$

$= 180V \pm 5V$

$= 180V \pm 2.8\%$ (Ans)



Example-4.2 Calculate the maximum percentage error in the difference of two voltages when $100V \pm 1\%$ and $V_2 = 80V \pm 5\%$.

Solution: $V_1 = 100V \pm 1\%$

$= 100V \pm 1V$

$V_2 = 80V \pm 5\%$

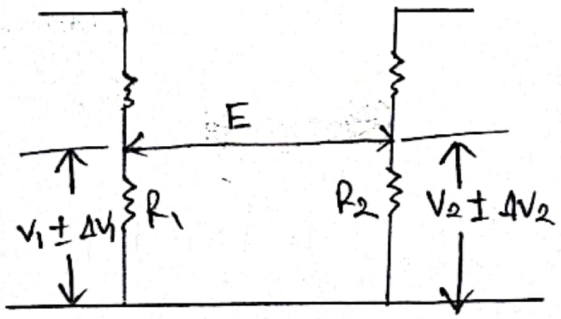
$= 80V \pm 4V$

$\therefore E = (V_1 - V_2) \pm (\Delta V_1 + \Delta V_2)$

$= (100 - 80) \pm (1 + 4)$

$= 20 \pm 5V$

$= 20 \pm 25\%$ (Ans)



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Thermocouple math

During experiments with a copper-constant thermocouple it was found that $c = 3.75 \times 10^{-2}$ mV/°C & $k = 4.5 \times 10^{-5}$ mV/°C². If $T_1 = 100$°C & the cold junction T_2 is kept in ice. Compute the e.m.f.

Solution: Here, $c = 3.75 \times 10^{-2}$ mV/°C

$k = 4.5 \times 10^{-5}$ mV/°C²

$T_1 = 100$°C

$T_2 = 0$°C (kept in ice)

We know,

$$E = c(T_1 - T_2) + k(T_1^2 - T_2^2)$$

$$= 3.75 \times 10^{-2}(100 - 0) + 4.5 \times 10^{-5}(100^2 - 0^2)$$

$$= (3.75 + 0.45)\ mV$$

$$= 4.20\ mV$$

(Ans)